

The z-Transform

Signals and Linear Systems

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- **Introduction:** Why the z-transform is the discrete-time Laplace transform
- **The z-Transform Definition**
- **The Region of Convergence (ROC):**
 - Right-sided, left-sided, and two-sided sequences
 - Poles, zeros, and the unit circle
 - ROC properties and significance
- **The Inverse z-Transform:**
 - Formal contour formula
 - Partial fractions and power-series inversion
- **Properties of the z-Transform**
- **LTI Systems, Causality, Stability, and Difference Equations**

Introduction and Definition

Continuous Time

Fourier: $e^{j\omega t}$ is an eigenfunction of LTI systems

$$x(t) = e^{j\omega t} \implies y(t) = H(j\omega) e^{j\omega t}$$

Laplace: generalize to $s = \sigma + j\omega$

$$x(t) = e^{st} \implies y(t) = H(s) e^{st}$$

Discrete Time

DTFT: $e^{j\omega n}$ is an eigenfunction of LTI systems

$$x[n] = e^{j\omega n} \implies y[n] = H(e^{j\omega}) e^{j\omega n}$$

z-Transform: generalize to $z = re^{j\omega}$

$$x[n] = z^n \implies y[n] = H(z) z^n$$

The Pattern

In both domains, Fourier analysis restricts us to the frequency axis. The broader transform (Laplace / z) allows a **complex** variable, letting us analyze unstable systems and characterize causality and stability.

The Eigenfunction Property of z^n

- For a discrete-time LTI system with impulse response $h[n]$, setting $x[n] = z^n$, the eigenvalue is the z-transform of $h[n]$:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] z^{n-k} = z^n \sum_{k=-\infty}^{\infty} h[k] z^{-k} = H(z) z^n.$$

The z-Transform

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

The z-transform is to the **DTFT** what the Laplace transform is to the **CTFT**.

What It Gives Us

- Extends analysis to a broader class of signals
- Lets us analyze **unstable** systems
- Reveals causality & stability via poles, zeros, and ROC

Geometric Difference

- s -plane: Fourier transform lives on the $j\omega$ -**axis**
- z -plane: DTFT lives on the **unit circle**

From Sampled Exponentials to z

Sampling e^{st} every T seconds gives

$$e^{s nT} = (e^{sT})^n$$

Define

$$z \triangleq e^{sT}$$

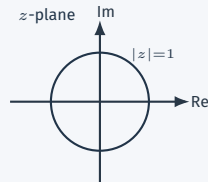
So the sampled signal is z^n .

The Change of Variable

$$z \triangleq e^{sT} = e^{\sigma T} e^{j\omega T} = r e^{j\theta}$$

$$|z| = r = e^{\sigma T} \leftrightarrow \text{damping}$$

$$\angle z = \theta = \omega T \leftrightarrow \text{frequency}$$



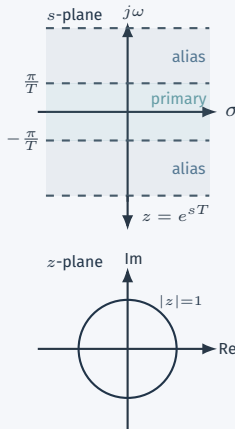
Why z instead of e^s ?

- In DTFT, we set $T = 1$ and work with $e^{j\omega n}$. We could generalize the same way and write e^{sn} , $s = \sigma + j\omega$.
- **But the map $s \mapsto e^{sT} = e^{\sigma T} e^{j\omega T}$ is many-to-one:** shifting ω by $\frac{2\pi}{T}$ gives the same $z = e^{sT}$.
- In continuous time, $e^{j(\omega+2\pi/T)t} \neq e^{j\omega t}$ for general t , so distinct s values still give distinct signals.
- In discrete time, n is an **integer**, so $e^{j(\omega+2\pi/T)Tn} = e^{j\omega Tn}$ —different s values produce the same discrete-time signal.
- Using z as a parameter solves this redundancy.

z Parameterization

Parameterizing by z collapses the redundancy in the s -plane. Also, as we will see, using z makes the algebra and geometry of the transform much more intuitive.

Many-to-one mapping



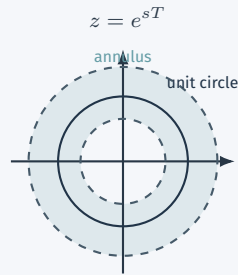
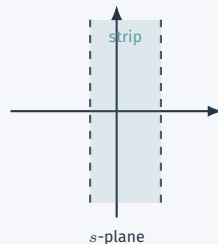
How the Mapping $z = e^{sT}$ Changes the Geometry

The mapping $z = e^{sT}$ transforms s -plane geometry into z -plane geometry:

s -plane	$\longrightarrow$	z -plane
Vertical line $\text{Re}\{s\} = \sigma_0$	$\longrightarrow$	Circle $ z = e^{\sigma_0 T}$
$j\omega$ -axis ($\sigma_0 = 0$)	$\longrightarrow$	Unit circle $ z = 1$
ROC strip	$\longrightarrow$	ROC annulus

Consequence

The unit circle plays the same role in the z -plane that the $j\omega$ -axis plays in the s -plane. ROC questions become questions about **radii**, not real parts.



The System Function $H(z)$

Since z^n is an eigenfunction of DT-LTI systems, the system is completely characterized by its **system function**:

System Function

$$H(z) = \sum_{k=-\infty}^{\infty} h[k] z^{-k} \qquad x[n] = z^n \implies y[n] = H(z) z^n$$

What $H(z)$ Encodes

- Convolution in time $\longrightarrow$ multiplication in z

$$Y(z) = H(z) X(z)$$

- Poles of $H(z)$ $\longrightarrow$ natural modes of the system
- Zeros of $H(z)$ $\longrightarrow$ frequencies the system blocks

Parallel with Laplace

Laplace	z-transform
$H(s)$	$H(z)$
$Y(s) = H(s)X(s)$	$Y(z) = H(z)X(z)$

Definition of the z-Transform and Its DTFT Connection

z-Transform

$$X(z) \triangleq \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Write

$$z = re^{j\omega}.$$

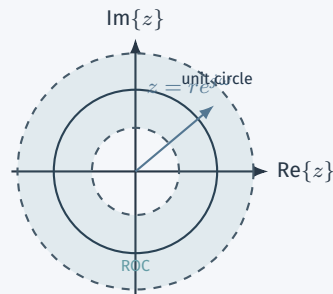
Then

$$\begin{aligned} X(re^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n](re^{j\omega})^{-n} \\ &= \sum_{n=-\infty}^{\infty} (x[n]r^{-n})e^{-j\omega n}. \end{aligned}$$

Hence

$$X(re^{j\omega}) = \mathcal{F}\{x[n]r^{-n}\}.$$

Setting $r = 1$ restricts the z-transform to the unit circle, so we recover the DTFT. Therefore the DTFT exists iff the ROC includes $|z| = 1$.



Fourier and z

The DTFT is the restriction of the z-transform to $|z| = 1$.

Examples and the ROC

Example 10.1: A Right-Sided Exponential

Consider

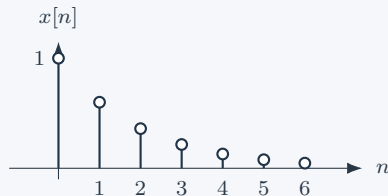
$$x[n] = a^n u[n].$$

Then

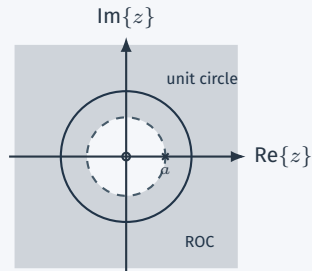
$$\begin{aligned} X(z) &= \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n, & |az^{-1}| < 1 \\ &= \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, & |z| > |a|. \end{aligned}$$

- This is the discrete-time analogue of $e^{-at}u(t)$ in the Laplace chapter.
- Pole at $z = a$, zero at $z = 0$.
- The ROC is the exterior of a circle.
- If $|a| < 1$, the ROC includes the unit circle, so the DTFT exists.

Sequence for $a = 0.6$



Pole-zero plot and ROC



Example 10.2: A Left-Sided Exponential

Consider

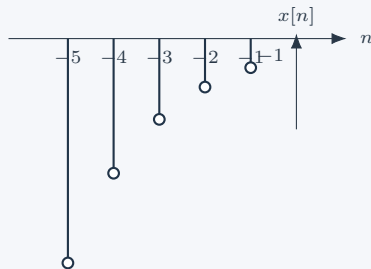
$$x[n] = -a^n u[-n - 1].$$

Then

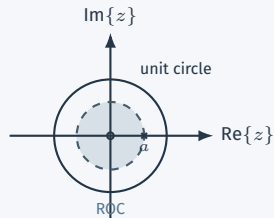
$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{-1} -a^n z^{-n} \\ &= - \sum_{m=1}^{\infty} a^{-m} z^m = - \sum_{m=1}^{\infty} (a^{-1} z)^m \\ &= \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \quad |z| < |a|. \end{aligned}$$

- Same algebraic expression as Example 10.1.
- But now the ROC is the interior of the circle.
- This is exactly the same message we saw with Laplace transforms: algebraic form alone is not enough.

Sequence for $a = 0.6$



Pole-zero plot and ROC



Why the ROC Matters

- Examples 10.1 and 10.2 have the same

$$X(z) = \frac{z}{z - a},$$

but they are different sequences because the ROCs differ.

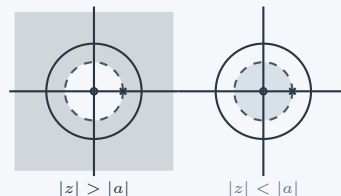
- So a z-transform is specified by

algebraic expression + ROC.

- For rational transforms:
 - zeros** are the roots of the numerator,
 - poles** are the roots of the denominator.
- The ROC tells us:
 - whether the sequence is right-sided, left-sided, or two-sided,
 - whether the DTFT exists,
 - and later, whether an LTI system is causal or stable.

Laplace: half-planes or strips $\iff$ z-transform: disks, exteriors, annuli

Same poles and zeros, different ROC



Example 10.3: Sum of Two Exponentials

$$x[n] = 7 \left(\frac{1}{3}\right)^n u[n] - 6 \left(\frac{1}{2}\right)^n u[n].$$

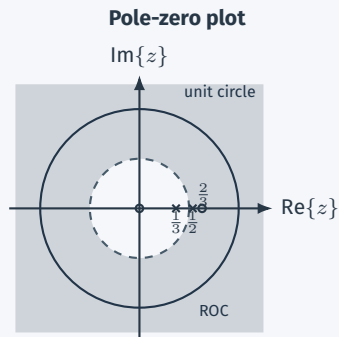
By linearity and Example 10.1,

$$\begin{aligned} X(z) &= \frac{7}{1 - \frac{1}{3}z^{-1}} - \frac{6}{1 - \frac{1}{2}z^{-1}} \\ &= \frac{1 - \frac{3}{2}z^{-1}}{\left(1 - \frac{1}{3}z^{-1}\right)\left(1 - \frac{1}{2}z^{-1}\right)} \\ &= \frac{z\left(z - \frac{2}{3}\right)}{\left(z - \frac{1}{3}\right)\left(z - \frac{1}{2}\right)}. \end{aligned}$$

Since both components are right-sided,

$$\text{ROC} : |z| > \frac{1}{2}.$$

- Poles at $z = \frac{1}{3}, \frac{1}{2}$.
- Zeros at $z = 0, \frac{2}{3}$.
- The ROC is outside the outermost pole, exactly as in the Laplace chapter.



Example 10.4: A Damped Sinusoid

Consider

$$x[n] = \left(\frac{1}{3}\right)^n \sin\left(\frac{\pi}{4}n\right) u[n].$$

Use Euler's identity:

$$\sin\left(\frac{\pi}{4}n\right) = \frac{e^{j\pi n/4} - e^{-j\pi n/4}}{2j}.$$

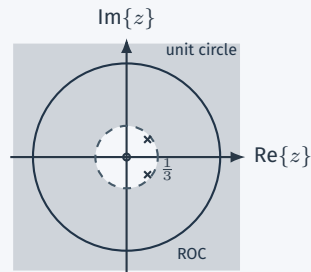
Then

$$\begin{aligned} X(z) &= \frac{1}{2j} \frac{1}{1 - \frac{1}{3}e^{j\pi/4}z^{-1}} - \frac{1}{2j} \frac{1}{1 - \frac{1}{3}e^{-j\pi/4}z^{-1}} \\ &= \frac{\frac{\sqrt{2}}{3}z^{-1}}{1 - \frac{\sqrt{2}}{3}z^{-1} + \frac{1}{9}z^{-2}} \\ &= \frac{\frac{\sqrt{2}}{3}z}{\left(z - \frac{1}{3}e^{j\pi/4}\right)\left(z - \frac{1}{3}e^{-j\pi/4}\right)}. \end{aligned}$$

Both exponential components are right-sided, so

$$\text{ROC} : |z| > \frac{1}{3}.$$

Sequence and ROC



Property 1

The ROC of $X(z)$ is a **ring centered at the origin**

$$X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]r^{-n}e^{-j\omega n}, \quad \sum_{n=-\infty}^{\infty} |x[n]|r^{-n} < \infty.$$

Absolute convergence depends on the radius $r = |z|$, not on the angle.

Property 2

The ROC cannot contain any poles.

At a pole, $X(z)$ is unbounded. Zeros may lie in the ROC; poles may only lie on its boundary or outside it.

Property 3

If $x[n]$ is finite-duration, the ROC is the entire z -plane, except possibly at $z = 0$ and/or $z = \infty$.

Justification: Only finitely many terms appear:

$$X(z) = \sum_{n=N_1}^{N_2} x[n]z^{-n}.$$

Thus every **finite, nonzero** z is safe.

But note there are two special cases:

- A sample at $n > 0$ creates negative powers of z and can exclude $z = 0$.
Since as $|z| \rightarrow 0$, $z^{-n} \rightarrow \infty$ for $n > 0$.
- A sample at $n < 0$ creates positive powers of z and can exclude $z = \infty$.
Since as $|z| \rightarrow \infty$, $z^{-n} \rightarrow \infty$ for $n < 0$.

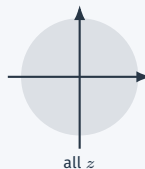
As a result, if $N_1 \geq 0$, the ROC includes ∞ ; if $N_2 \leq 0$, it includes 0.

Example 10.5: Finite-Duration Sequences

Impulse

$$\delta[n] \xleftrightarrow{\mathcal{Z}} \sum_{n=-\infty}^{\infty} \delta[n] z^{-n} = 1.$$

ROC: entire z-plane, including 0 and ∞ .



Delayed impulse

$$\delta[n-1] \xleftrightarrow{\mathcal{Z}} \sum_{n=-\infty}^{\infty} \delta[n-1] z^{-n} = z^{-1}.$$

ROC: entire z-plane except $z = 0$.



Advanced impulse

$$\delta[n+1] \xleftrightarrow{\mathcal{Z}} z$$

ROC: entire finite z-plane, but not $z = \infty$.



These examples show how positive powers of z threaten ∞ and negative powers threaten 0.

Property 4

If $x[n]$ is a right-sided sequence, and if the circle $|z| = r_0$ is in the ROC, then all finite values of z for which $|z| > r_0$ will also be in the ROC.

Right-sided means $x[n] = 0$ for $n < N_1$, so

$$X(z) = \sum_{n=N_1}^{\infty} x[n]z^{-n}, \quad \sum_{n=N_1}^{\infty} |x[n]|r_0^{-n} < \infty$$

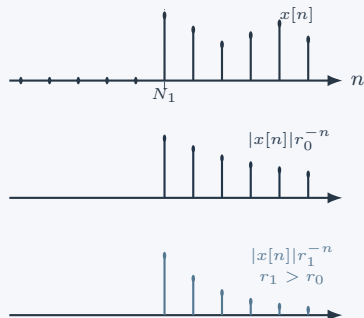
when $|z| = r_0$ lies in the ROC. For any $r_1 > r_0$:

$$|x[n]|r_1^{-n} = |x[n]|r_0^{-n} \left(\frac{r_0}{r_1} \right)^n.$$

Since $0 < r_0/r_1 < 1$, the extra factor decays with $n \Rightarrow$ absolute summability at r_1 is guaranteed.

Endpoint nuance: if $N_1 < 0$, ∞ may be excluded; for causal sequences ($N_1 \geq 0$), $\infty \in \text{ROC}$.

Faster decay for $r_1 > r_0$



Property 5

If $x[n]$ is a left-sided sequence, and if the circle $|z| = r_0$ is in the ROC, then all values of z for which $0 < |z| < r_0$ will also be in the ROC.

Left-sided means $x[n] = 0$ for $n > N_2$, so

$$X(z) = \sum_{n=-\infty}^{N_2} x[n]z^{-n}, \quad \sum_{n=-\infty}^{N_2} |x[n]|r_0^{-n} < \infty$$

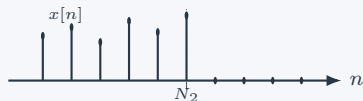
when $|z| = r_0$ lies in the ROC. For any $0 < r_1 < r_0$:

$$|x[n]|r_1^{-n} = |x[n]|r_0^{-n} \left(\frac{r_0}{r_1}\right)^n.$$

Since $r_0/r_1 > 1$ but $n \rightarrow -\infty$, the extra factor decays $\Rightarrow$ absolute summability at r_1 is guaranteed.

Endpoint nuance: if $N_2 > 0$, 0 may be excluded; for anticausal sequences ($N_2 \leq 0$), $0 \in \text{ROC}$.

Faster decay for $0 < r_1 < r_0$



Property 6

If $x[n]$ is two-sided and has a z-transform, the ROC is an annulus.

Split a two-sided sequence into

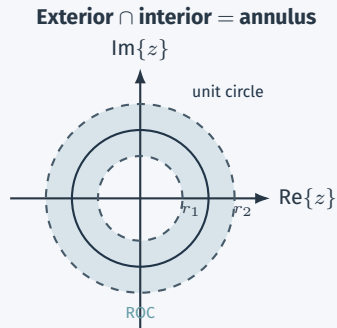
$$x[n] = x[n]u[n - N] + x[n](1 - u[n - N]).$$

The right-sided part contributes an exterior ROC, and the left-sided part contributes an interior ROC.

Overlap test

The z-transform exists only if those two ROCs overlap. When they do, the overlap is a ring:

$$r_1 < |z| < r_2.$$



Example 10.6: A Finite-Length Exponential

For

$$x[n] = \begin{cases} a^n, & 0 \leq n \leq N-1, a > 0 \\ 0, & \text{otherwise} \end{cases}$$

we have

$$\begin{aligned} X(z) &= \sum_{n=0}^{N-1} a^n z^{-n} = \frac{1 - (az^{-1})^N}{1 - az^{-1}} \\ &= \frac{z [1 - a^N z^{-N}]}{z - a}. \end{aligned}$$

Since the sequence is finite-duration, the ROC is all z except possibly $z = 0$.

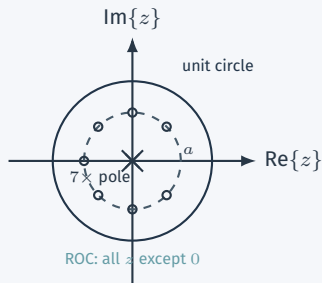
After canceling the factor at $z = a$, the zeros are

$$z_k = ae^{j2\pi k/N}, \quad k = 1, 2, \dots, N-1,$$

and there are $N-1$ poles at the origin.

This is a beautiful discrete-time picture: the zeros lie evenly on a circle of radius a .

Pole-zero pattern for $N = 8$



Example 10.7: $b^{|n|}$

Let

$$x[n] = b^{|n|}, \quad b > 0.$$

Write it as

$$x[n] = b^n u[n] + b^{-n} u[-n-1].$$

Then the ROC depends strongly on b :

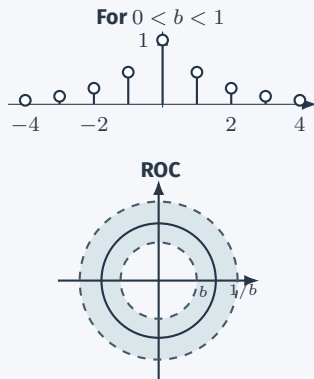
- If $0 < b < 1$, the right-sided part needs $|z| > b$ and the left-sided part needs $|z| < 1/b$. Hence

$$b < |z| < \frac{1}{b}.$$

- If $b > 1$, the required conditions become $|z| > b$ and $|z| < 1/b$, which cannot both hold.
- Therefore **no ROC exists** for $b > 1$.

Important lesson

A formal algebraic manipulation can exist, but without a nonempty ROC there is no z-transform.



Property 7

If $X(z)$ is rational, the ROC is bounded by poles or extends all the way to 0 and/or ∞ .

Property 8

If $X(z)$ is rational and $x[n]$ is right-sided, the ROC is outside the outermost pole.

More precisely: the ROC is outside the circle through the finite pole of largest magnitude. If the sequence is causal, the ROC also includes $z = \infty$.

Property 9

If $X(z)$ is rational and $x[n]$ is left-sided, the ROC is inside the innermost **nonzero** pole.

More precisely: it extends inward and may include $z = 0$. If the sequence is anticausal, the ROC includes $z = 0$.

Same Algebra / Different Signals

Poles draw the possible circular boundaries. The ROC chooses the side: exterior $\Rightarrow$ right-sided, interior $\Rightarrow$ left-sided, annulus $\Rightarrow$ two-sided.

Example 10.8: Three Possible ROCs

Consider

$$X(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})}.$$

The poles are at

$$z = \frac{1}{2}, \quad z = 2.$$

Hence there are three possible ROCs:

$$|z| > 2, \quad \frac{1}{2} < |z| < 2, \quad |z| < \frac{1}{2}.$$

- Exterior ROC: right-sided sequence.
- Interior ROC: left-sided sequence.
- Middle annulus: two-sided sequence.
- Only the middle annulus includes the unit circle, so only that one has a DTFT.



$$|z| > 2$$



$$\frac{1}{2} < |z| < 2$$



$$|z| < \frac{1}{2}$$

The Inverse z-Transform

Recall

$$X(re^{j\omega}) = \mathcal{F}\{x[n]r^{-n}\}.$$

If the circle $|z| = r$ lies in the ROC, apply the inverse DTFT:

$$x[n]r^{-n} = \frac{1}{2\pi} \int_{2\pi} X(re^{j\omega})e^{j\omega n} d\omega.$$

Multiply by r^n :

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(re^{j\omega})(re^{j\omega})^n d\omega.$$

With $z = re^{j\omega}$ and $dz = jre^{j\omega} d\omega = jz d\omega$, we obtain

$$x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1} dz$$

where C is any counterclockwise closed contour lying entirely within the ROC.

Inverse z-Transform

$$x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1} dz$$

Example 10.9: Partial Fractions with an Exterior ROC

Consider

$$X(z) = \frac{3 - \frac{5}{2}z^{-1}}{\left(1 - \frac{1}{4}z^{-1}\right)\left(1 - \frac{1}{3}z^{-1}\right)}, \quad |z| > \frac{1}{3}.$$

Partial fractions give

$$X(z) = \frac{1}{1 - \frac{1}{4}z^{-1}} + \frac{2}{1 - \frac{1}{3}z^{-1}}.$$

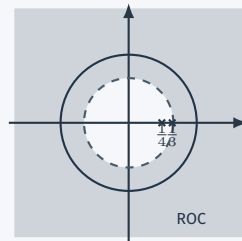
Because the ROC is outside the outermost pole, both terms are right-sided:

$$\frac{1}{1 - \frac{1}{4}z^{-1}} \xleftrightarrow{\mathcal{Z}} \left(\frac{1}{4}\right)^n u[n],$$

$$\frac{2}{1 - \frac{1}{3}z^{-1}} \xleftrightarrow{\mathcal{Z}} 2 \left(\frac{1}{3}\right)^n u[n].$$

Hence

$$x[n] = \left(\frac{1}{4}\right)^n u[n] + 2 \left(\frac{1}{3}\right)^n u[n].$$



Main idea

The partial fractions do not change.
The **ROC decides how each term is interpreted.**

Examples 10.10 and 10.11: The Same Algebraic $X(z)$ Gives Different Sequences

Keep the same partial-fraction expansion as Example 10.9:

$$X(z) = \frac{1}{1 - \frac{1}{4}z^{-1}} + \frac{2}{1 - \frac{1}{3}z^{-1}}.$$

Example 10.10

ROC:

$$\frac{1}{4} < |z| < \frac{1}{3}.$$

Then the first term is right-sided but the second is left-sided:

$$x[n] = \left(\frac{1}{4}\right)^n u[n] - 2\left(\frac{1}{3}\right)^n u[-n - 1].$$



Example 10.11

ROC:

$$|z| < \frac{1}{4}.$$

Now both terms are left-sided:

$$x[n] = -\left(\frac{1}{4}\right)^n u[-n - 1] - 2\left(\frac{1}{3}\right)^n u[-n - 1].$$



Intuition

Rewrite $X(z)$ as a power series that is valid in the ROC, then match it with

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}.$$

The coefficient of z^{-n} is $x[n]$; positive powers of z therefore represent negative-time samples.

- Example 10.12**

$$X(z) = 4z^2 + 2 + 3z^{-1} \implies x[n] = 4\delta[n+2] + 2\delta[n] + 3\delta[n-1].$$

- Example 10.13**

$$X(z) = \frac{1}{1 - az^{-1}}.$$

If $|z| > |a|$ ($|az^{-1}| < 1$),

$$X(z) = 1 + az^{-1} + a^2z^{-2} + \dots \implies x[n] = a^n u[n].$$

If $|z| < |a|$ ($|z/a| < 1$),

$$X(z) = -\frac{z/a}{1 - z/a} = -a^{-1}z - a^{-2}z^2 - a^{-3}z^3 - \dots \implies x[n] = -a^n u[-n-1].$$

Consider

$$X(z) = \log(1 + az^{-1}), \quad |z| > |a|.$$

From the geometric series expansion, we have:

$$\frac{1}{1+t} = 1 - t + t^2 - t^3 + \dots, \quad |t| < 1.$$

Integrating from 0 to v gives

$$\log(1+v) = \int_0^v \frac{1}{1+t} dt = \int_0^v (1 - t + t^2 - \dots) dt = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} v^n}{n}, \quad |v| < 1.$$

Now set $v = az^{-1}$. Since $|z| > |a|$, we have $|az^{-1}| < 1$, so

$$X(z) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} a^n z^{-n}}{n}.$$

Matching powers of z^{-n} gives

$$x[n] = \begin{cases} (-1)^{n+1} \frac{a^n}{n}, & n \geq 1 \\ 0, & n \leq 0. \end{cases}$$

Why power series help

The coefficient of z^{-n} in the expansion is exactly the sequence value $x[n]$.

Properties of the z-Transform

Linearity

If

$$x_1[n] \xleftrightarrow{\mathcal{Z}} X_1(z), \quad x_2[n] \xleftrightarrow{\mathcal{Z}} X_2(z),$$

then

$$ax_1[n] + bx_2[n] \xleftrightarrow{\mathcal{Z}} aX_1(z) + bX_2(z)$$

with ROC containing at least $R_1 \cap R_2$.

Justification: the defining sum is linear. The ROC is at least the intersection; it can be larger if pole-zero cancellation occurs.

Example

$$7 \left(\frac{1}{3}\right)^n u[n] - 6 \left(\frac{1}{2}\right)^n u[n]$$

is Example 10.3.

Time Shifting

If

$$x[n] \xleftrightarrow{\mathcal{Z}} X(z),$$

then

$$x[n - n_0] \xleftrightarrow{\mathcal{Z}} z^{-n_0} X(z)$$

with the same ROC, except possibly for adding or deleting 0 or ∞ .

Proof sketch:

$$\sum_{n=-\infty}^{\infty} x[n - n_0] z^{-n} = z^{-n_0} \sum_{m=-\infty}^{\infty} x[m] z^{-m}.$$

The multiplier z^{-n_0} can add/delete a pole at 0 or at ∞ , but it does not move any finite nonzero boundary.

Example

$$\delta[n - 1] \xleftrightarrow{\mathcal{Z}} z^{-1}.$$

Scaling in the z-domain

$$a_0^n x[n] \xleftrightarrow{\mathcal{Z}} X\left(\frac{z}{a_0}\right)$$

and the ROC scales by $|a_0|$.

Proof:

$$\sum a_0^n x[n] z^{-n} = \sum x[n] \left(\frac{z}{a_0}\right)^{-n}.$$

Multiplying by $e^{j\omega_0 n}$ rotates the pole-zero pattern; multiplying by r_0^n scales its radii.

Time Reversal

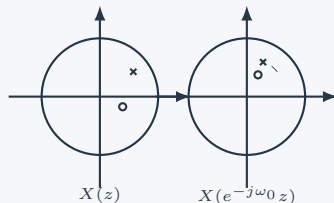
$$x[-n] \xleftrightarrow{\mathcal{Z}} X(z^{-1})$$

with inverted ROC.

Proof:

$$\sum x[-n] z^{-n} = \sum_m x[m] (z^{-1})^{-m}.$$

Effect of multiplying by $e^{j\omega_0 n}$



Conjugation

$$x^*[n] \xleftrightarrow{\mathcal{Z}} X^*(z^*)$$

with the same ROC.

$$\sum x^*[n] z^{-n} = \left(\sum x[n] (z^*)^{-n} \right)^* = X^*(z^*).$$

For real $x[n]$, $X(z) = X^*(z^*)$, so nonreal poles and zeros occur in conjugate pairs and the ROC is symmetric about the real axis.

Define the expanded sequence

$$x_{(k)}[n] = \begin{cases} x[n/k], & n \text{ is an integer multiple of } k, \\ 0, & n \text{ is not a multiple of } k. \end{cases}$$

Time expansion property

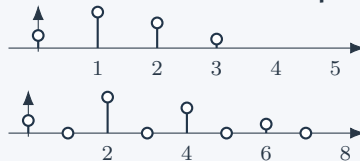
$$x_{(k)}[n] \xleftrightarrow{\mathcal{Z}} X(z^k),$$

with ROC $R^{1/k}$.

Proof from the series:

$$\sum_{n=-\infty}^{\infty} x_{(k)}[n] z^{-n} = \sum_{m=-\infty}^{\infty} x[m] z^{-km} = X(z^k).$$

Insert $k - 1$ zeros between samples



ROC Geometry

z is in the new ROC exactly when z^k is in the old ROC, so radii are mapped by $r \mapsto r^k$.

Convolution

$$y[n] = x_1[n] * x_2[n] \xleftrightarrow{\mathcal{Z}} Y(z) = X_1(z)X_2(z)$$

with ROC containing at least $R_1 \cap R_2$; cancellation may enlarge it.

Starting from

$$y[n] = \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k],$$

we obtain

$$\begin{aligned} Y(z) &= \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k]z^{-n} \\ &= \sum_{k=-\infty}^{\infty} x_1[k]z^{-k} \sum_{m=-\infty}^{\infty} x_2[m]z^{-m} \quad (m = n - k) \\ &= X_1(z)X_2(z). \end{aligned}$$

ROC Nuance

Convergence is guaranteed wherever both series converge, so the ROC includes at least $R_1 \cap R_2$. If a pole is cancelled in the product, the final ROC can be larger.

First Difference: Example 10.15

For $h[n] = \delta[n] - \delta[n - 1]$,

$$y[n] = x[n] - x[n - 1] \xleftrightarrow{\mathcal{Z}} (1 - z^{-1})X(z),$$

with ROC at least R except possible deletion of $z = 0$; cancellation may add $z = 1$.

First difference is the discrete-time counterpart of differentiation.

Accumulation

Example 10.16 defines the running sum

$$w[n] = \sum_{k=-\infty}^n x[k].$$

Since

$$w[n] = u[n] * x[n],$$

and

$$u[n] \xleftrightarrow{\mathcal{Z}} \frac{1}{1 - z^{-1}}, \quad |z| > 1,$$

we get

$$W(z) = \frac{1}{1 - z^{-1}} X(z).$$

ROC contains at least $R \cap \{|z| > 1\}$.

Intuition: convolution with $u[n]$ is the discrete-time counterpart of integration.

Differentiation in the z-domain

$$n x[n] \xleftrightarrow{\mathcal{Z}} -z \frac{dX(z)}{dz}$$

with $\text{ROC} = R$.

Proof:

$$\frac{dX}{dz} = \sum_n (-n) x[n] z^{-n-1}; \quad -z \frac{dX}{dz} = \sum_n n x[n] z^{-n}.$$

Example 10.18 Starting from

$$a^n u[n] \xleftrightarrow{\mathcal{Z}} \frac{1}{1 - az^{-1}},$$

differentiation gives

$$n a^n u[n] \xleftrightarrow{\mathcal{Z}} \frac{az^{-1}}{(1 - az^{-1})^2}.$$

Linear constant-coefficient difference equation

If

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k],$$

then, using linearity and time shifting,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}}.$$

Justification:

$$\mathcal{Z}\{y[n-k]\} = z^{-k}Y(z), \quad \mathcal{Z}\{x[n-k]\} = z^{-k}X(z).$$

Then factor $Y(z)$ on the left and $X(z)$ on the right.

Book warning

The difference equation determines the **algebraic** system function, but not the ROC. The ROC is extra information: it decides causality, stability, and the impulse response.

Summary of z-Transform Properties

Property	Time-domain signal	z-domain expression	ROC / nuance
Linearity	$ax_1[n] + bx_2[n]$	$aX_1(z) + bX_2(z)$	$\supseteq R_1 \cap R_2$
Time shift	$x[n - n_0]$	$z^{-n_0} X(z)$	same R , except $0, \infty$
Scaling	$a_0^n x[n]$	$X(z/a_0)$	$ a_0 R$
Time reversal	$x[-n]$	$X(z^{-1})$	inverted R
Conjugation	$x^*[n]$	$X^*(z^*)$	same R
Time expansion	$x_{(k)}[n]$: insert $k - 1$ zeros	$X(z^k)$	$R^{1/k}$
Convolution	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	$\supseteq R_1 \cap R_2$
First difference	$x[n] - x[n - 1]$	$(1 - z^{-1})X(z)$	possible 0 deletion, 1 cancellation
Accumulation	$\sum_{k=-\infty}^n x[k]$	$\frac{1}{1-z^{-1}} X(z)$	$\supseteq R \cap \{ z > 1\}$
z-domain differentiation	$nx[n]$	$-z dX(z)/dz$	same R
Initial value	causal/right-sided from 0	$x[0] = \lim_{z \rightarrow \infty} X(z)$	requires $x[n] = 0, n < 0$
LCCDE	$\sum a_k y[n - k] = \sum b_k x[n - k]$	$H(z) = \frac{\sum b_k z^{-k}}{\sum a_k z^{-k}}$	equation alone does not set ROC

Analysis of LTI Systems

Transform-Domain View of an LTI System

- For an LTI system with impulse response $h[n]$,

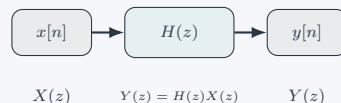
$$Y(z) = H(z)X(z).$$

- The z-transform converts convolution into multiplication.
- If the ROC of $H(z)$ includes the unit circle, then

$$H(e^{j\omega})$$

is the **frequency response**.

- So poles, zeros, and the ROC describe not only the algebra of the transform, but also the behavior of the system.



Eigenfunction Reminder

For input $x[n] = z^n$,

$$y[n] = H(z)z^n.$$

Main payoff

Just as with Laplace transforms, many system properties become visible directly in the transform domain.

Causality Test

A discrete-time LTI system is causal iff the ROC of $H(z)$ is the exterior of a circle and includes ∞ .

Example 10.20

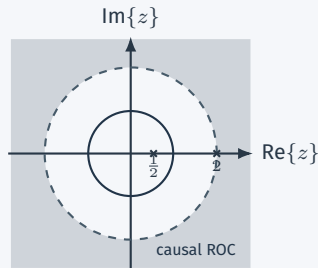
$$H(z) = \frac{z^3 - 2z^2 + z}{z^2 + \frac{1}{4}z + \frac{1}{8}}$$

is **not causal** because the numerator degree exceeds the denominator degree when written in powers of z .

Example 10.21

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1 - 2z^{-1}}, \quad |z| > 2.$$

The ROC is outside the outermost pole and includes ∞ , so the system is causal.



Stability Test

An LTI system is stable iff the ROC of $H(z)$ includes the unit circle.

- **Example 10.22:** the same algebraic $H(z)$ from Example 10.21 is unstable if the ROC is $|z| > 2$.
- **Example 10.23:** for the causal first-order system

$$H(z) = \frac{1}{1 - az^{-1}},$$

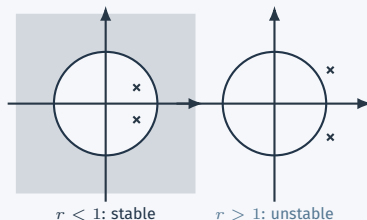
stability requires $|a| < 1$.

- **Example 10.24:** if

$$H(z) = \frac{1}{1 - 2r \cos \theta z^{-1} + r^2 z^{-2}},$$

then the poles are at $re^{\pm j\theta}$, so a causal system is stable iff $r < 1$.

Complex poles and the unit circle



Example 10.25: Difference Equations Become Algebra

Consider the linear constant-coefficient difference equation

$$y[n] - \frac{1}{2}y[n-1] = x[n] + \frac{1}{3}x[n-1].$$

Taking z-transforms and using linearity and time shifting,

$$Y(z) - \frac{1}{2}z^{-1}Y(z) = X(z) + \frac{1}{3}z^{-1}X(z).$$

So

$$Y(z) = X(z) \frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

and therefore

$$H(z) \triangleq \frac{Y(z)}{X(z)} = \frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{1}{2}z^{-1}}.$$

The difference equation fixes the algebraic form of $H(z)$, but not the ROC:

$$|z| > \frac{1}{2} \text{ for the causal interpretation,} \quad |z| < \frac{1}{2} \text{ for the anticausal one.}$$

Main lesson: a difference equation determines the algebraic form of $H(z)$, but the ROC still determines whether the realization is causal, anticausal, or stable.

Solving a Difference Equation by z-Transform

Use the **causal** system from Example 10.25:

$$H(z) = \frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}.$$

Let the input be the unit step:

$$x[n] = u[n] \xleftrightarrow{\mathcal{Z}} X(z) = \frac{1}{1 - z^{-1}}, \quad |z| > 1.$$

Then

$$Y(z) = H(z)X(z) = \frac{1 + \frac{1}{3}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - z^{-1})}.$$

Partial fractions give

$$Y(z) = \frac{8/3}{1 - z^{-1}} - \frac{5/3}{1 - \frac{1}{2}z^{-1}}.$$

Hence

$$y[n] = \frac{8}{3}u[n] - \frac{5}{3}\left(\frac{1}{2}\right)^n u[n].$$

This is the discrete-time version of what we did with differential equations in the Laplace domain:

difference equation $\implies$ algebra $\implies$ inverse transform.